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EX NO: 8B

SHIELDING EFFECTIVENESS AND EMC COMPLIANCE ANALYSIS USING MATLAB

DATE: 11/08/26

OBJECTIVES

- i) To simulate the shielding effectiveness (SE) of conductive shielding materials using MATLAB.
 - ii) To analyze the effect of reflection loss, absorption loss, and multiple reflection loss on shielding effectiveness.
 - iii) To evaluate EMC compliance by comparing measured EMI levels with standard EMC emission limits using MATLAB.
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OBJECTIVE 1

SIMULATION OF SHIELDING EFFECTIVENESS

PROCEDURE

1. Open MATLAB and create a new script file.
 2. Define the operating frequency range.
 3. Define shielding material properties.
 4. Calculate the skin depth.
 5. Calculate the absorption loss.
 6. Determine the shielding effectiveness.
 7. Plot shielding effectiveness versus frequency.
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FORMULA

Skin Depth

$$\delta = \sqrt{2 / \sigma \omega \mu}$$

Absorption Loss

$$A = 8.686 (t / \delta)$$

Shielding Effectiveness

$$SE = R + A + B$$

Where:

- **R** = Reflection Loss
 - **A** = Absorption Loss
 - **B** = Multiple Reflection Loss
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RESULT

The shielding effectiveness of the conductive enclosure was successfully simulated.

OBJECTIVE 2

ANALYSIS OF REFLECTION LOSS, ABSORPTION LOSS AND MULTIPLE REFLECTION LOSS

PROCEDURE

1. Calculate reflection loss.
 2. Calculate absorption loss.
 3. Calculate multiple reflection loss.
 4. Determine total shielding effectiveness.
 5. Plot individual loss components.
 6. Compare their contributions.
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FORMULA

Reflection Loss

$$R = 168 - 10 \log_{10}(\sigma_r / \mu_r f)$$

Absorption Loss

$$A = 131.4 t \sqrt{f \mu_r \sigma_r}$$

Total Shielding Effectiveness

$$SE = R + A + B$$

RESULT

The contributions of reflection, absorption, and multiple reflection losses were analyzed successfully.

OBJECTIVE 3

EMC COMPLIANCE ANALYSIS

PROCEDURE

1. Define EMC emission limit values.
 2. Import measured EMI data.
 3. Compare measured emissions with EMC limits.
 4. Identify non-compliant frequencies.
 5. Plot the EMI spectrum and EMC limit line.
 6. Generate a compliance report.
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FORMULA

Compliance Margin

$$\text{Margin} = \text{Limit} - \text{Emission}$$

If:

Margin > 0 → Compliant

Margin < 0 → Non-Compliant

RESULT

EMC compliance was evaluated successfully using MATLAB.

OVERALL MATLAB PROGRAM

```
clc;
clear;
close all;

%% Objective 1 : Shielding Effectiveness Simulation

f = logspace(6,9,1000);    % 1 MHz to 1 GHz
mu0 = 4*pi*1e-7;
sigma = 5.8e7;             % Copper conductivity
t = 1e-3;                  % Shield thickness
omega = 2*pi*f;
```

```

delta = sqrt(2./(omega*mu0*sigma));
A = 8.686*(t./delta);    % Absorption Loss (dB)
%% Objective 2 : Reflection and Multiple Reflection Loss
sigma_r = 1;
mu_r = 1;
R = 168 - 10*log10((sigma_r)./(mu_r*f));
B = 5;    % Approximate Multiple Reflection Loss
SE= R + A + B;
%% Objective 3 : EMC Compliance Analysis
EMC_Limit = 50*ones(size(f));
Measured_Emission = ...
    40 + 5*sin(2*pi*log10(f)/3);
Margin = EMC_Limit - Measured_Emission;
Compliance = Margin > 0;
%% Display Results
fprintf('\nSHIELDING EFFECTIVENESS RESULTS\n');
fprintf('-----\n');
fprintf('Maximum SE = %.2f dB\n',max(SE));
fprintf('Minimum SE = %.2f dB\n',min(SE));
if all(Compliance)
    fprintf('System is EMC COMPLIANT\n');
else
    fprintf('System is NOT EMC COMPLIANT\n');
end
%% Plot Results
figure('Name', ...
    'Shielding Effectiveness and EMC Compliance', ...
    'NumberTitle','off');
subplot(2,2,1);
semilogx(f,A,'LineWidth',2);

```

```
grid on;
title('Absorption Loss');
xlabel('Frequency (Hz)');
ylabel('Loss (dB)');
subplot(2,2,2);
semilogx(f,R,'LineWidth',2);

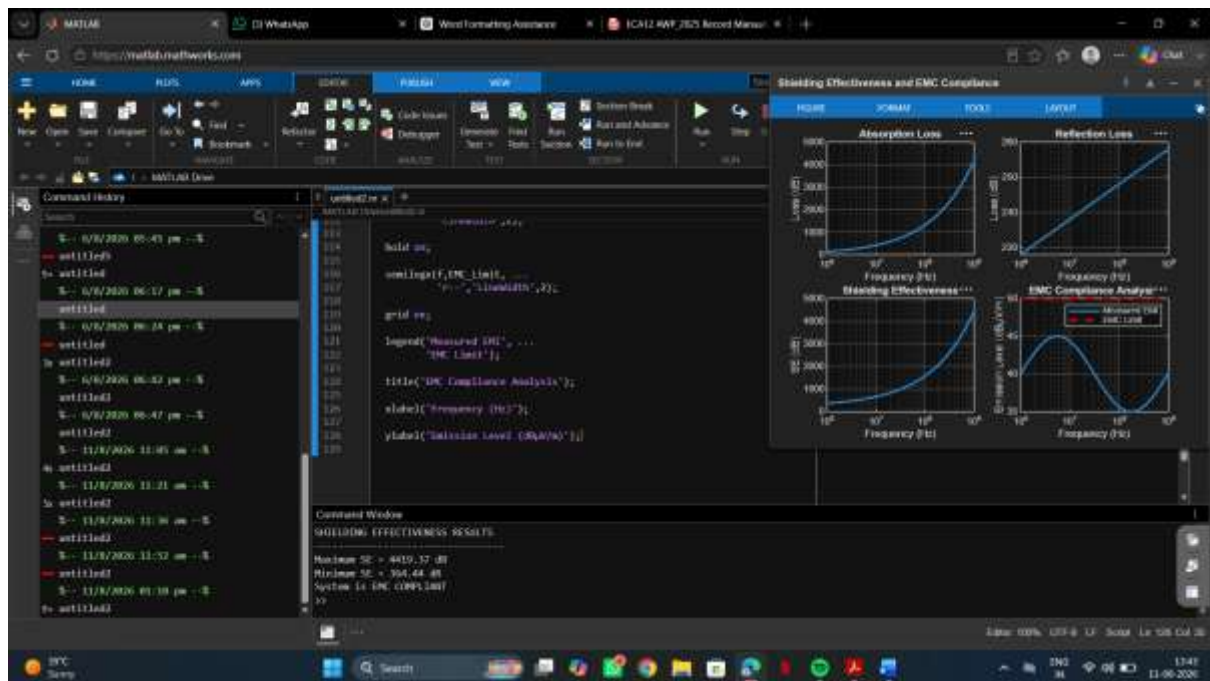
grid on;
title('Reflection Loss');
xlabel('Frequency (Hz)');
ylabel('Loss (dB)');
subplot(2,2,3);
semilogx(f,SE,'LineWidth',2);

grid on;
title('Shielding Effectiveness');
xlabel('Frequency (Hz)');
ylabel('SE (dB)');
subplot(2,2,4);
semilogx(f,Measured_Emission,'LineWidth',2);

hold on;
semilogx(f,EMC_Limit,'r--','LineWidth',2);

grid on;
legend('Measured EMI','EMC Limit');
title('EMC Compliance Analysis');
xlabel('Frequency (Hz)');
ylabel('Emission Level (dBμV/m)');
```

OUTPUT :



FINAL RESULT

The shielding effectiveness of the conductive shielding material was successfully simulated using MATLAB. The reflection loss, absorption loss, and multiple reflection loss were analyzed, and the total shielding effectiveness was determined. The EMC compliance analysis was also successfully performed by comparing the measured EMI emission levels with the specified EMC limit.